



# Testing (in) The Clouds

*Ashraf Gaffar*

# SW Complexity

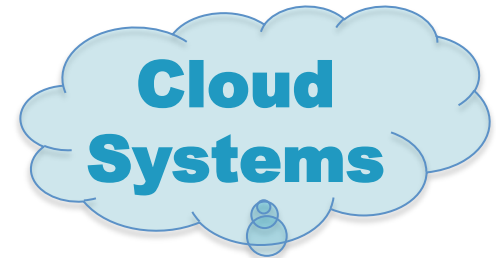
- UNIX: 4501 LoC (3976 Kernel + 374 Shell)
- MS-DOS: 13,587
- Linux: 27mi
- 95-98-XP: 15-18-35 mi
- Windows 7: 40mi
- MacOs: 100-200 mi

## Web Systems

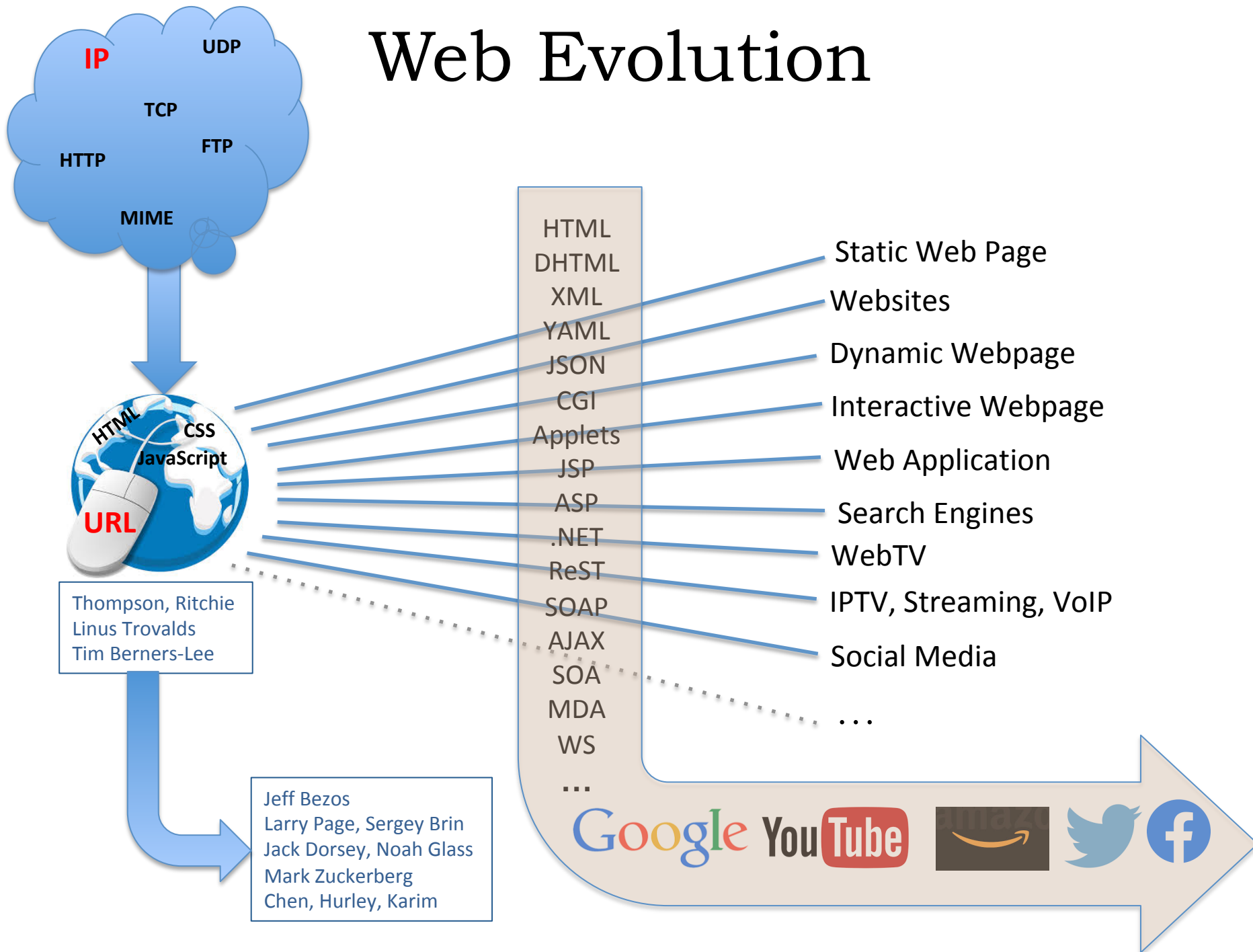
- Google: 2 Bi
- The Genome Project: 3 Bi

## Embedded Systems

- Space Shuttle: 400 K (0.4mi)
- Modern car: 100mi
- Tesla: 150mi

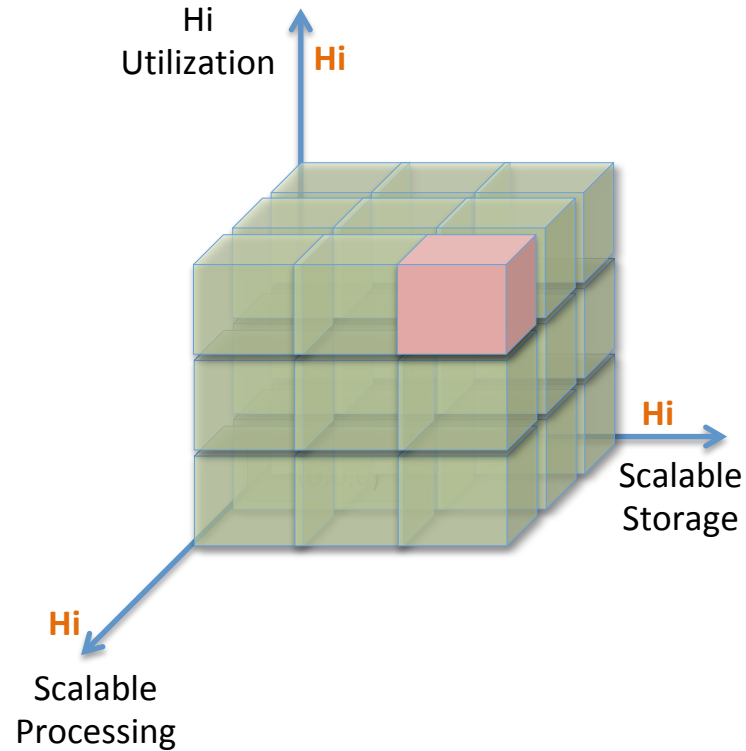


# Web Evolution





# Cloud



SaaS

PaaS

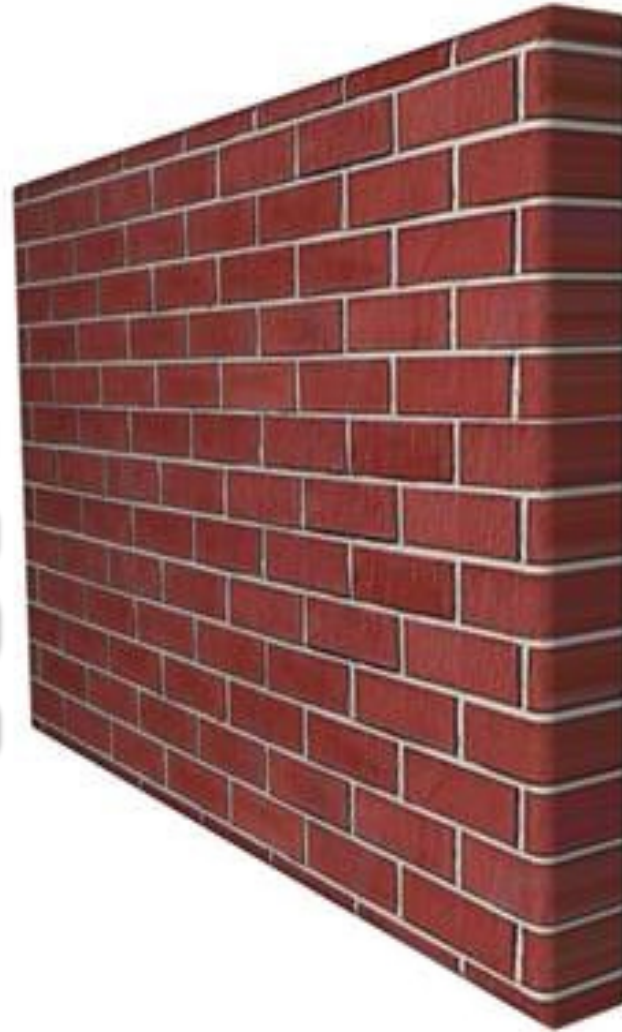
IaaS

Serverless Architecture: The Web is Yours



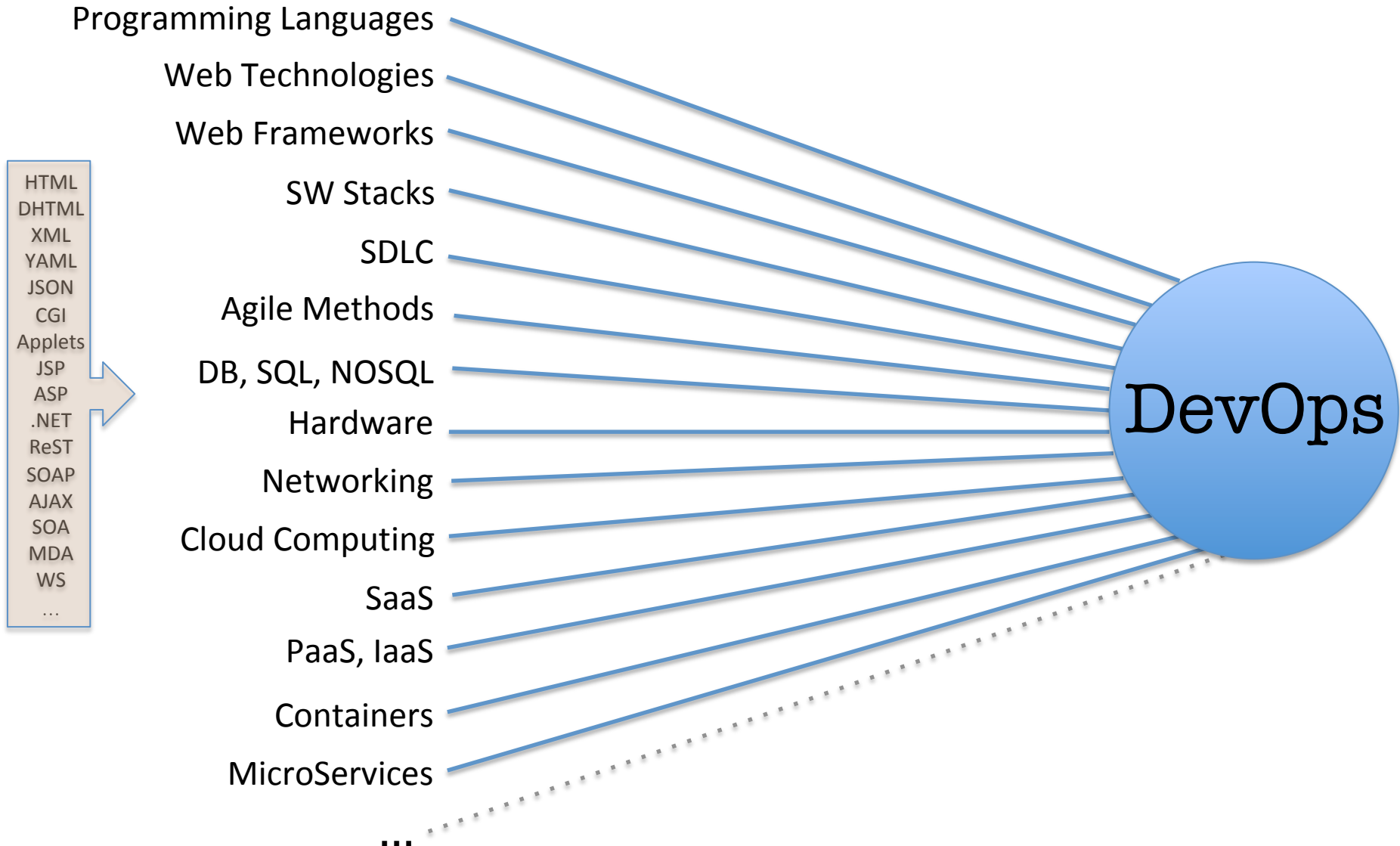
# Software Engineering in Industry

UR  
Analysis, Synthesis  
Requirement  
Design (**PT**, **UX**, **Usability**)  
Development (+**Des.** **QA**)  
Testing, Q-Gates  
Deployment, Ramp-up  
Handover



**IMS**

# Ecosystem



# DevOps

Integrated,  
Automated,  
High Throughput  
Workflow

With

Fast Delivery of  
High Quality,  
Low Risk,  
Low Cost  
Software

Revolution  
Symbiosis  
Synergy  
Stability  
Self-Repair  
Robustness  
World System  
Community  
Collective Success  
Umbrella  
Culture  
New Era  
Major Leap Forward

## DevOps Defies Definition



# DevOps Software Engineering

## SE

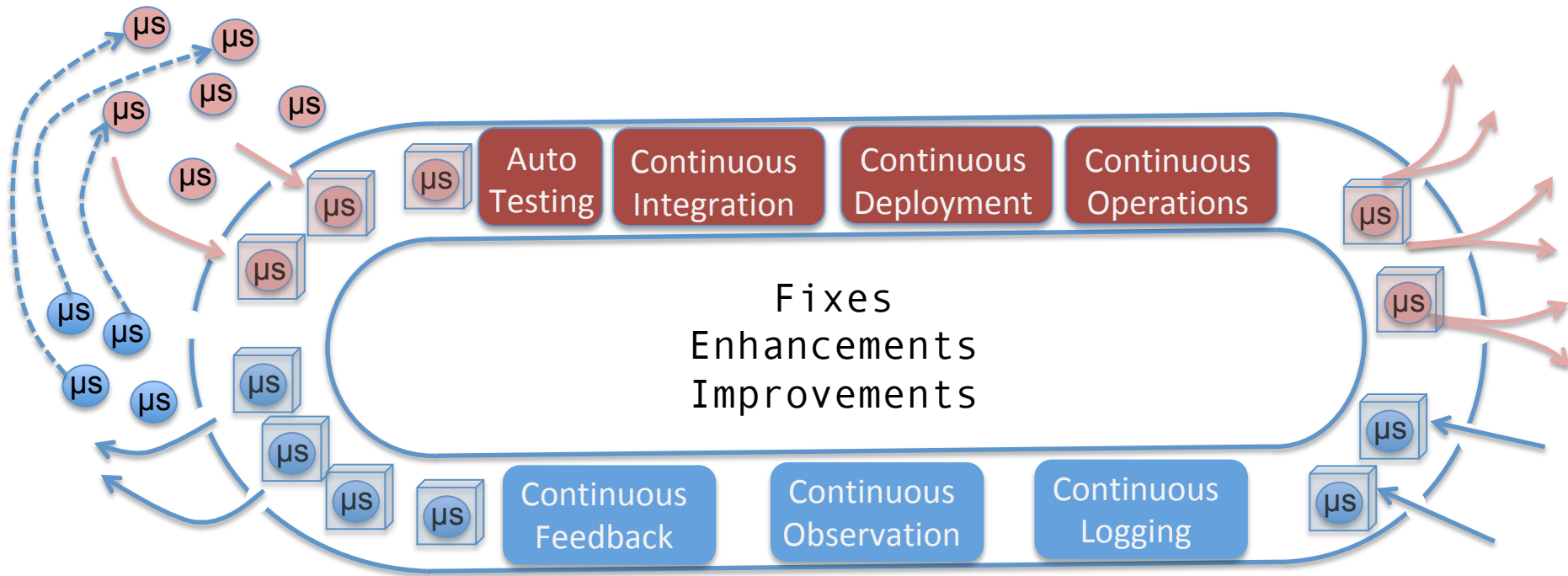
- SW Life Cycles, Agile Scrum
- Software Testing (Unit, Integ., Sys)
- UCD, Usability, BPM, IMS, UX
- Web technologies & Frameworks
- SW Architecture, MDA, SOA, WS
- OOP, Event Handling, EDP
- Synchronous, Asynchronous, ORB
- DB, RDBMS
- O/S, Kernel, System Calls, VM

## SE 2.0

- Kanban
- Test automation, CI, CD
- Ops
- Cloud, Containers
- Micro Services
- Orchestration, Choriography
- Auto Deployment
- NOSQL
- Containers

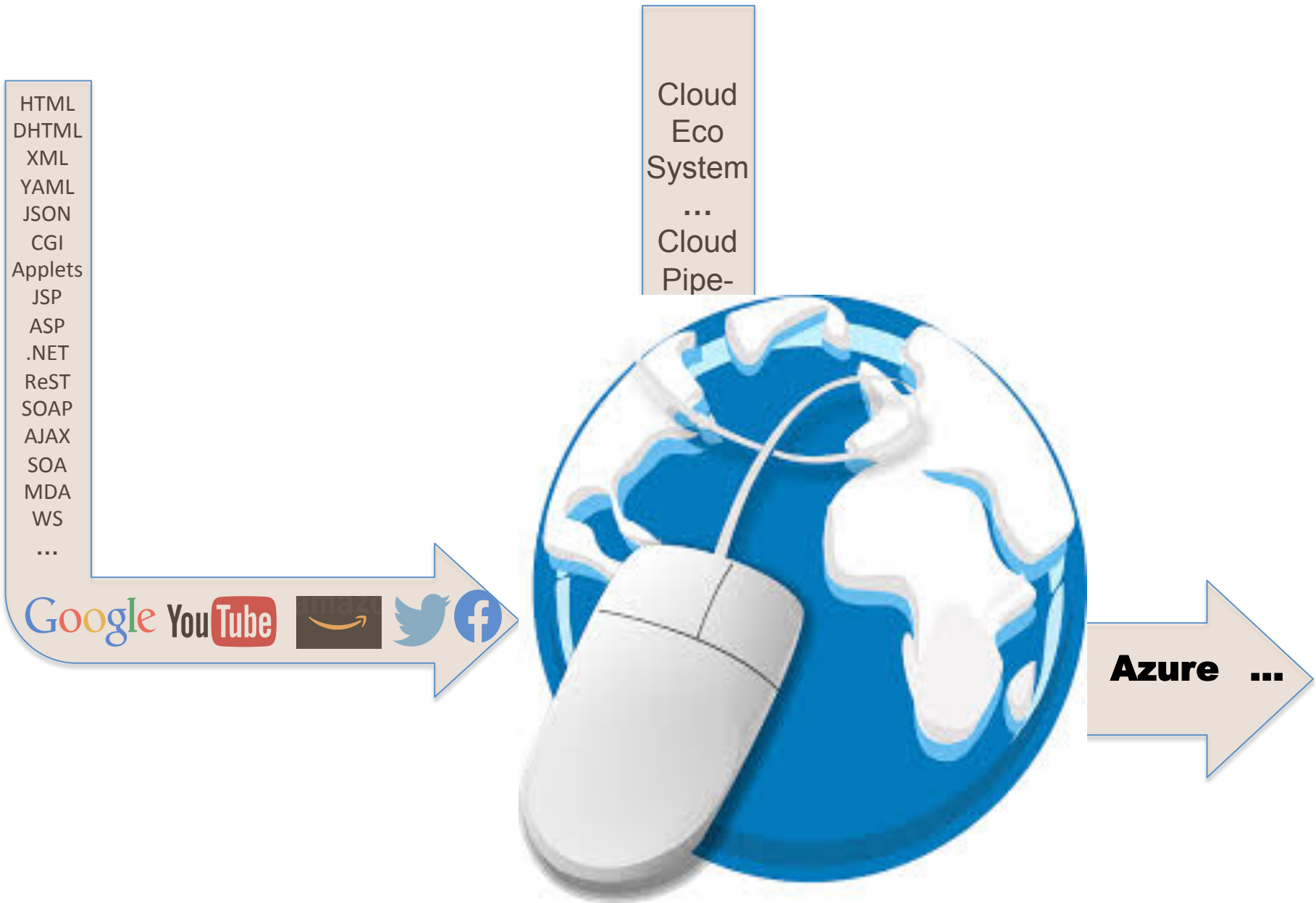
# Dev

# Ops

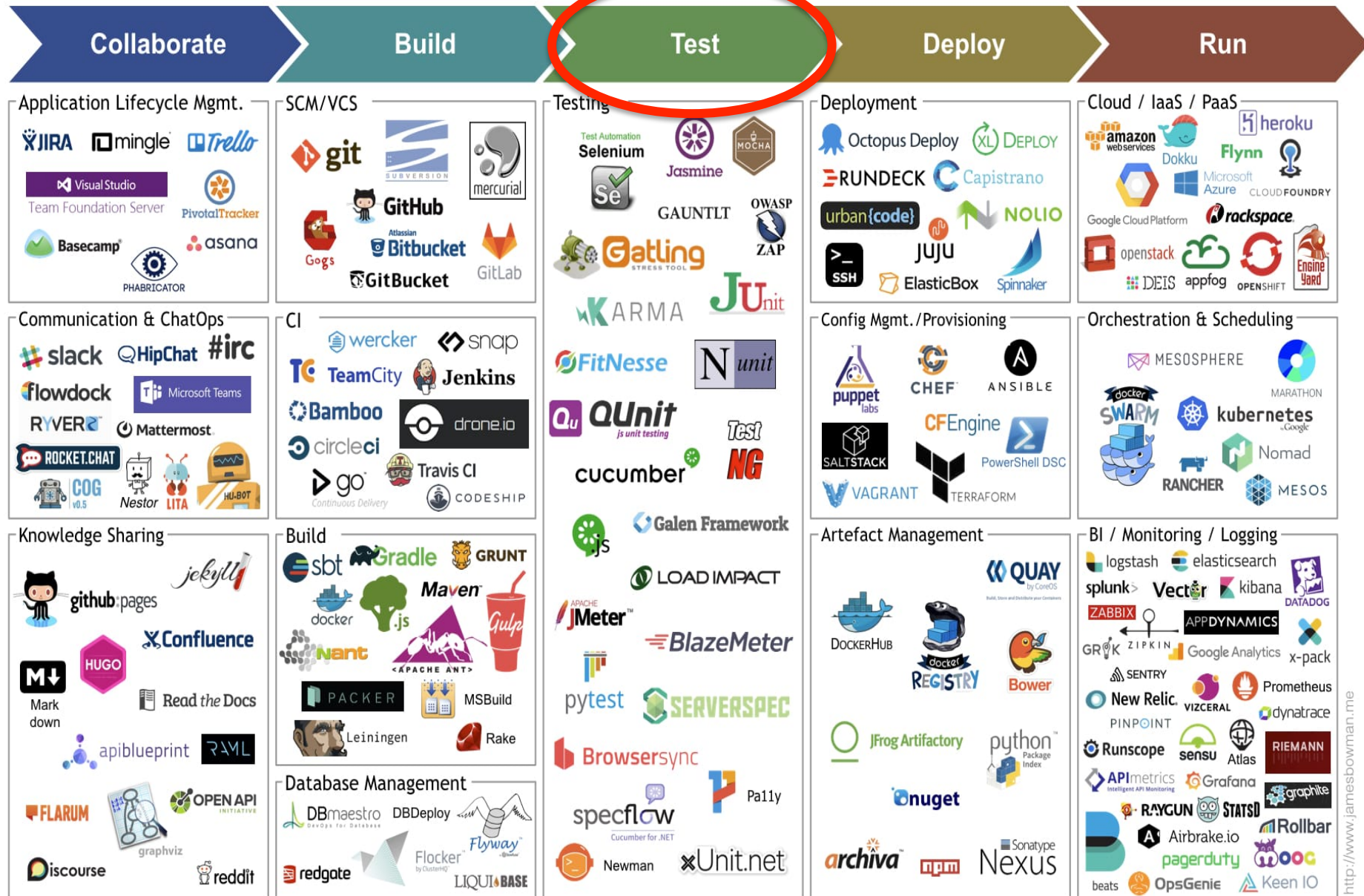


Continuous pipeline of automated tools in an ecosystem spanning technology and business

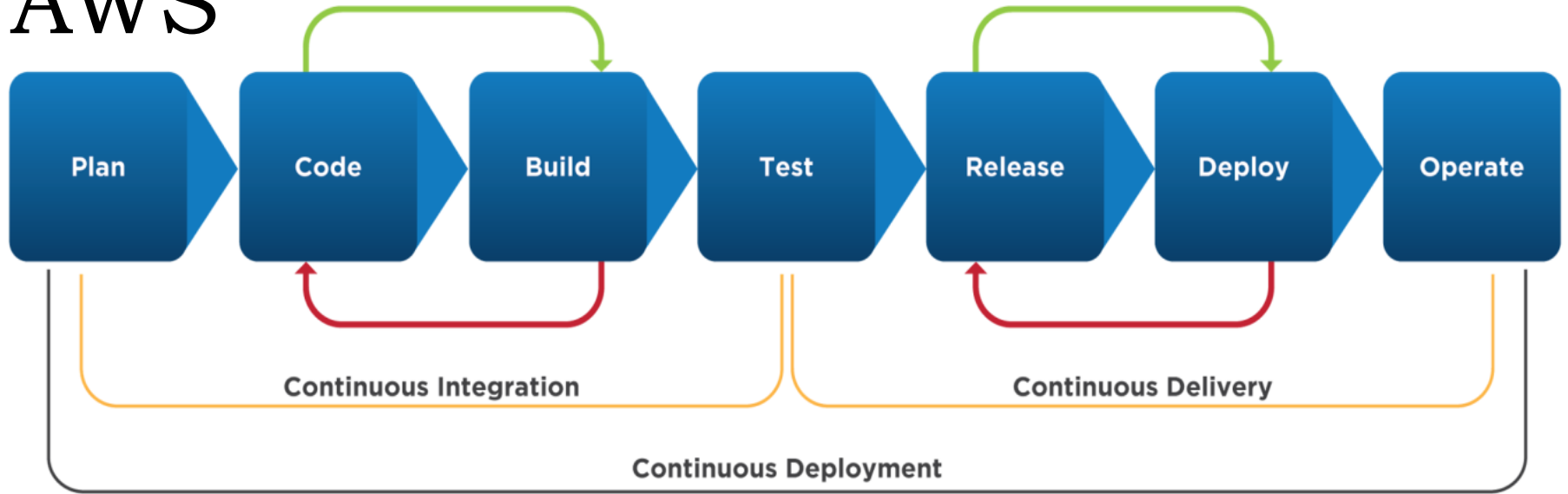
# Cloud Evolution



# Pipeline Automation Tools



# AWS



## Planning

- Requirement finalization
- Updates & new changes
- Architecture & design
- Task assignment
- Timeline finalization

## Code

- Development
- Configuration finalization
- Check-in source code
- Static-code analysis
- Automated review & peer review

## Build

- Compile code
- Unit testing
- Code-metrics
- Build container images or package
- Preparation or update in deployment templated
- Create or update monitor dashboards

## Test

- Integration test with other component
- Load & stress test
- UI testing
- Penetration testing
- Requirement testing

## Release

- Preparing release notes
- Version tagging
- Code freeze
- Feature freeze

## Deploy

- Updating the infrastructure i.e staging, production
- Verification on deployment i.e smoke tests

## Operate

- Monitor designed dashboard
- Alarm triggers
- Automatic critical events handler
- Monitor error logs



AWS CodeCommit



AWS CodePipeline



Terraform



Travis CI



CHEF



AWS Elastic Beanstalk



AWS CodeDeploy



AWS Lambda



AWS X-Ray



AWS CloudTrail



AWS CloudFormation



AWS OpsWorks



AWS ECS



AWS Config



AWS CloudWatch

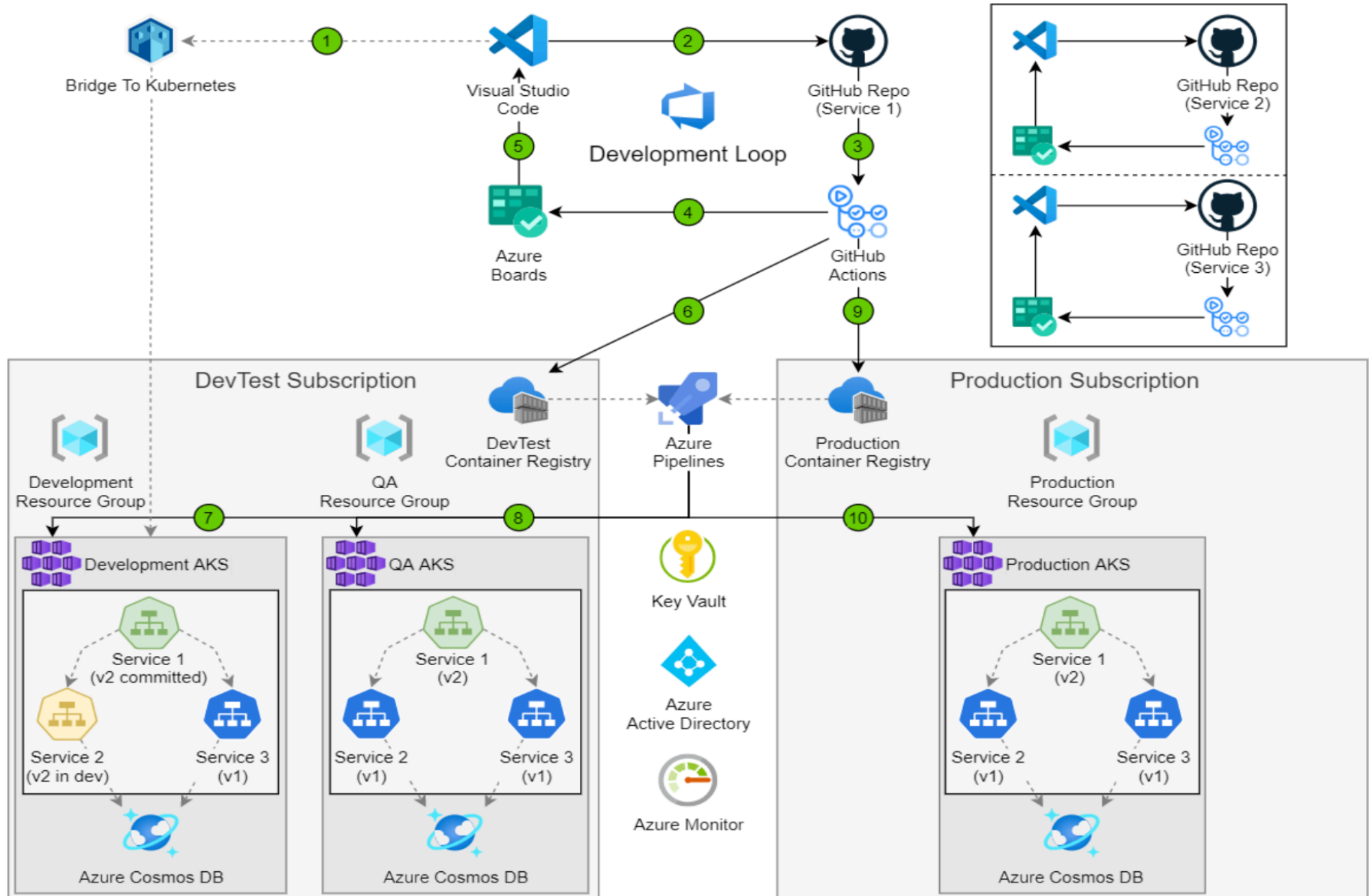




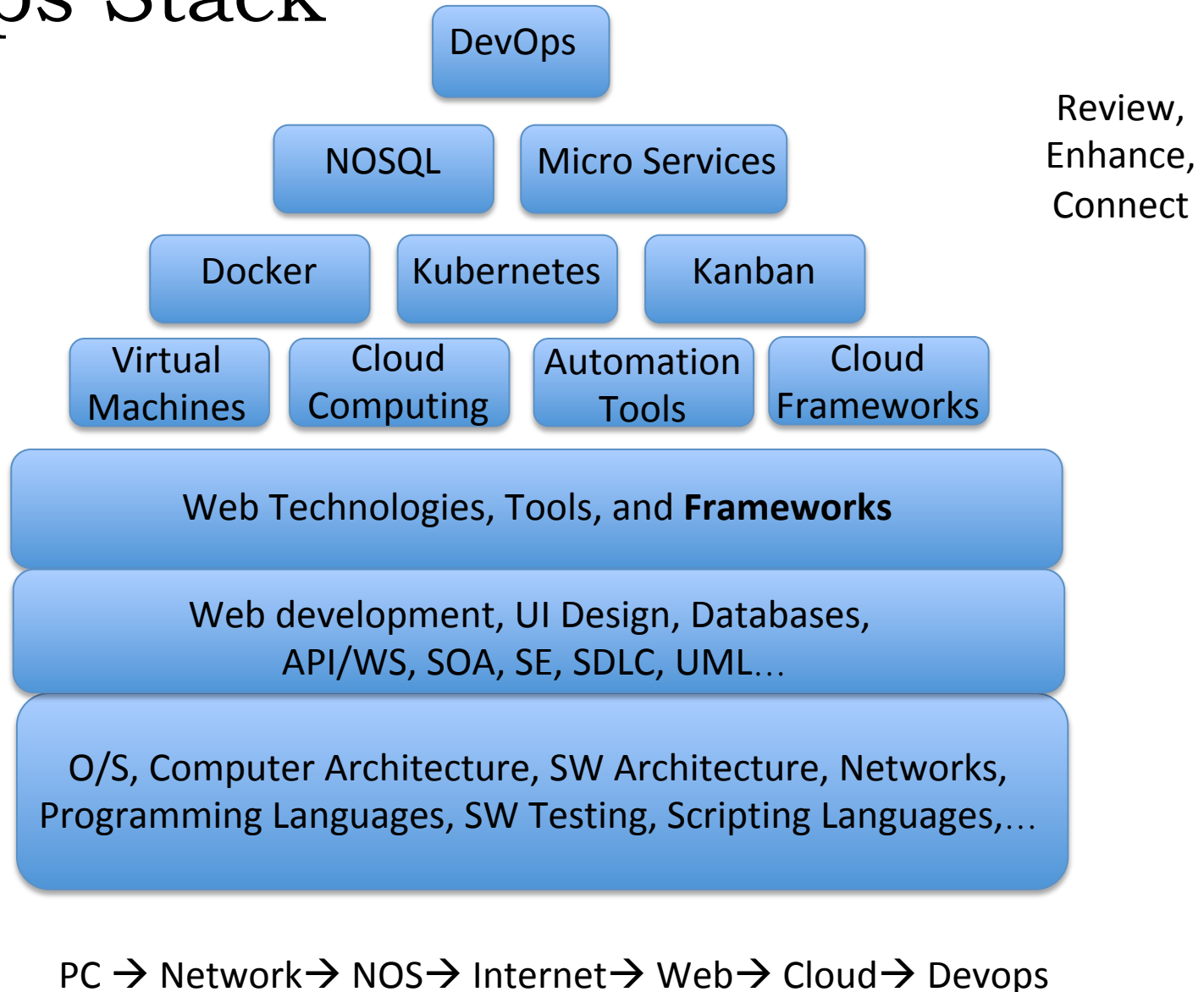
Users

Users

# Azure Workflow



# DevOps Stack



# Phase 1- Cloud Evaluation (Resource Comparison)

- Evaluate before testing
- Comprehensive evaluation
- Several areas involved
- Extended Bandwidth
  - Pre-testing
    - SaaS, PaaS, IaaS
    - CSP
  - Cloud Testing
    - Cloud architectures
    - Software architectures

# Cloud Landscape Evaluation

- Landscape Survey
- Cloud Native Vs. Cloud Agnostic Decision
  - Tools
  - Compatibility
  - Standards and Automation
- Pricing
  - Design time
  - Development time
  - Operations
    - Low, average, maximum
- Disaster Recovery



# Phase 2- Initial Cloud Testing

- Performance, Load, and Stress Testing
  - Theoretical
- SLA
- QA
- Security
- Privacy

# Cloud Tools & SW Prototypes (Shallow)

- Multiple incremental PTs
- Configurability and versioning
  - Much more complex
- Tool assessment
  - Mostly for Run Time (Operations)
  - But also for Design & Build time
  - Test Tools
- Mixing & matching CSP
- Transitioning
  - On premises to hybrid to full cloud

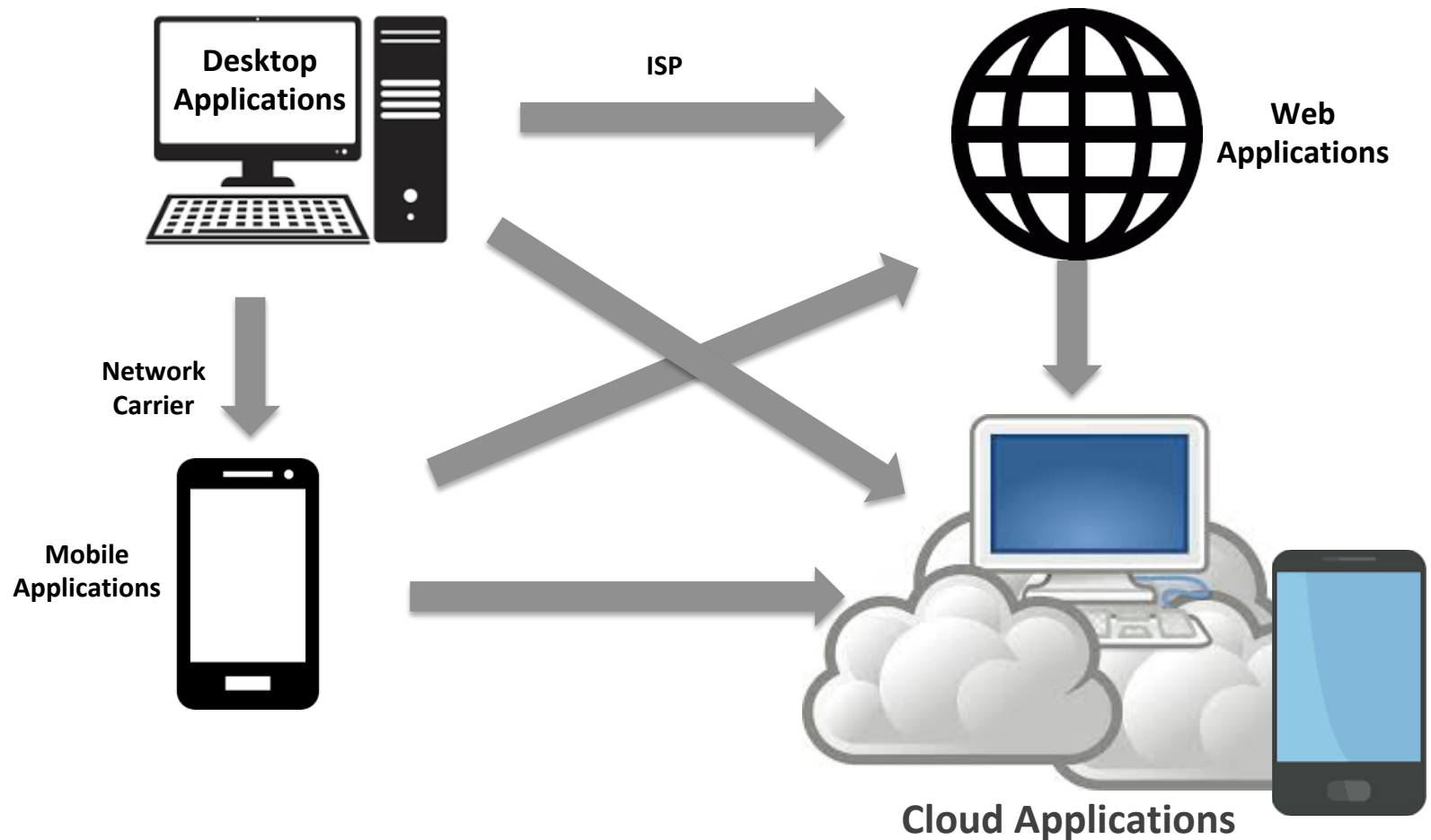
# Phase III: Full Prototype (Deep & Wide) Testing

- Select and build the development (**design & build time**) environment
- Create the user (**runtime**) environment
- Build one full vertical stack (D) with rich functionality (W)
- **Run** traditional SW functional testing
  - Unit, integration & system testing
  - Smoke testing
  - Sanity testing
  - White- & black box testing
  - Broad system testing
- **Pair** with non-functional testing on the cloud (**next →** )

# Non-Functional (Operations) Testing

- More critical than traditional SW testing
- Much harder on the cloud
  - Black box, Elasticity, Scalability
- Storage, processing, and bandwidth testing
- Cost testing
- Interoperability testing
- Latency testing
- Performance testing
- Browser compatibility testing

# Cloud Test Architecture



# Types of Tests

## ◆ White Box Testing

- Code Analysis
- Path Analysis

## ◆ Black Box Testing

- Boundary Value Analysis
- Equivalence Partitioning
- Decision Tables
- State Transition (Graph)

## ◆ Hybrid Testing

- A Mix of white and black testing



# Software Testing Lifecycle (STLC)

Integrated with SDLC

## 1. Analysis

1. Requirements and/or code

## 2. Test Planning

## 3. Test case design

## 4. Test preparation: Environment setup

## 5. Execution & Logging

## 6. Test result analysis

# Test Planning

## 1. Scope Definition

- Features included, excluded, and dependencies

## 2. Goals & Objectives

## 3. Test Aspects

- Functionality, usability, performance, reliability, security

## 4. Approach

- Test types to be performed

## 5. Schedule & Budget

## 6. Test Plan Approval

# Hard but Critical Cloud Testing

- Availability testing
- Load testing
- Security testing
- Multi-tenancy testing
- Stress & recovery testing

# TaaS

- Industry today: Lack of experience in cloud testing
  - Different skill set required: Cloud Engineers
- Lack of standards
- Lack of staff
- Enormity of landscape
- TaaS is a viable option
  - Cloud performance
  - Cloud security
  - Software functionality

# Cloud Operations Bonus Testing

- Advanced simulation
- Data migration
- Legacy system integration
- Live updates
- Live upgrades
- Compliance/GRC
- Scalability
- Cost analysis and optimization
- Continuous security assessment and monitoring